

Anomaly Detection to Detect Data Anomalies

1. Overview

Traditional monitoring uses fixed thresholds:

None

Alert if CPU > 90%

Alert if Errors > 5%

But many real-world issues are not fixed-threshold problems.

Examples:

- Orders suddenly drop 40% at midnight
- Traffic doubles only in one city
- Revenue unusually low for Sunday
- Duplicate payments spike slowly

These are best detected using **Anomaly Detection**.

Anomaly Detection identifies unusual behavior compared to normal patterns.

2. What is an Anomaly?

An anomaly is a datapoint or pattern that significantly deviates from expected behavior.

Examples:

- Sudden spike
- Sudden drop
- Seasonal mismatch
- Unexpected trend shift
- Rare category burst

3. Why It Matters in HLD

Used for:

Reliability

- Error spikes
- Latency anomalies
- Queue lag changes

Data Quality

- Missing ETL rows
- Duplicate records
- Abnormal null values

Business Analytics

- Revenue drop
- Churn increase
- Conversion crash

Security

- Login abuse
- Fraud transactions
- Suspicious traffic bursts

4. Example Scenarios

Example 1: E-commerce

Normal Sunday sales:

None

₹10 lakh

Today:

None

₹4 lakh

Anomaly may indicate:

- Payment issue
- Checkout bug
- Market event

Example 2: API Errors

Typical errors:

None
0.2%

Suddenly:

None
2.7%

Likely deployment issue.

Example 3: ETL Pipeline

Expected rows per hour:

None
50,000

Today:

None
31,000

Possible missing data.

5. How Basic Anomaly Detection Works

Step 1: Collect Historical Data

Examples:

- Orders/hour
- Latency/min
- Revenue/day

Step 2: Training Set

Model learns normal behavior.

Step 3: Validation Set

Tune model parameters.

Step 4: Test Set

Measure performance before production.

Step 5: Live Scoring

Incoming datapoint gets anomaly probability.

None

`Metric → Model → Score → Alert if abnormal`

6. Common Types of Anomalies

Point Anomaly

Single strange datapoint.

None

`Traffic = 10x normal`

Contextual Anomaly

Depends on context.

None

`Low traffic at noon = abnormal`

`Low traffic at 3AM = normal`

Collective Anomaly

Sequence pattern abnormal.

None

`Gradual memory rise for 3 hours`

7. Simpler Alternatives (Often Used in Industry)

Not all systems need ML.

Moving Average

Compare current vs recent average.

Seasonal Baseline

Compare this Monday 9AM vs previous Mondays 9AM.

Standard Deviation Bands

Alert if outside expected range.

Percent Change Threshold

None

Revenue drops > 30%

8. Precision vs Recall

Precision

When alert fires, how often correct?

High precision = fewer false alarms.

Recall

How many real anomalies detected?

High recall = fewer misses.

Tradeoff exists.

9. Operational Challenges

Important to mention.

- False positives (noise)
- False negatives (missed incidents)
- Concept drift (behavior changes over time)
- Seasonality handling
- Cold start (little historical data)
- Labeling data expensive
- Explainability difficult

10. HLD Architecture

None

Metrics / Events Stream



Feature Processing



Anomaly Detection Engine



Alerting System



Dashboard / Investigation

11. Critical Dataset Use Cases

Reserved for high-value domains:

- Payments
- Fraud detection
- Inventory mismatch
- Revenue dashboards
- Ad click fraud
- Security monitoring

12. Security Example

None

Normal login attempts/min = 200

Today = 8,000 from one ASN

Anomaly may indicate attack.

13. Business Example

None

Cart adds normal

Checkout completions collapse

Likely payment outage or UX bug.

14. Common Tools

- Python (Scikit-learn)
- Spark ML
- TensorFlow
- Prometheus anomaly plugins
- Datadog Watchdog
- New Relic AI monitoring

15. Interview Script

Say:

“For dynamic metrics where static thresholds fail, I’d use anomaly detection based on historical baselines and seasonality. It helps detect silent issues like revenue drops, ETL misses, or unusual latency spikes.”

Excellent answer.

16. Final Summary

- Anomaly detection finds unusual patterns
- Better than static thresholds for dynamic systems
- Useful for ops, analytics, fraud, data quality
- Needs historical data + tuning
- Expensive, so use for critical metrics first

17. Memorable Line

Thresholds catch the obvious. Anomaly detection catches the unexpected.